

How It Works

- Trains an autoencoder on **normal traffic only**
- Normal traffic → low reconstruction error
- Attacks → high reconstruction error (anomaly)
- Threshold-based classification: [error > threshold → anomaly](#)

2. VAE Experiments & Deployment

2.1. VAE Training & Evaluation

- **Goal:** Replace original Autoencoder with Variational Autoencoder (VAE) to improve FPR.
- **Actions:**
 - Trained VAE ($\beta=1.0$) on normal data.
 - Evaluated on benchmark test set:
 - FPR: 75.61% (improved from 83.77% of original AE)
 - Recall: 98.73%
- **Further Experiments:**
 - Trained $\beta=2.0$ and $\beta=4.0$ variants.
 - $\beta=2.0$: FPR 78.69%, Recall 99.09%
 - $\beta=4.0$: FPR 75.99%, Recall 95.79%
 - **Conclusion:** $\beta=1.0$ VAE is optimal (lowest FPR with recall $\geq 95\%$).

2.2. VAE Integration

- **Code Changes:**
 - [autoencoder.py](#): Added [reduction](#) parameter to VAE's [reconstruction_error](#).
 - [agent_one.py](#): Added VAE import, `model_type` parameter for auto-detection.
 - [test_accuracy_verification.py](#): Updated to load VAE model with threshold 0.027361.
 - **Results:** VAE deployed in Agent One, accuracy test run.
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3. Threshold Recalibration for Raw Data

3.1. Calibration Script

- **Goal:** Optimize threshold for raw UNSW-NB15 data (different distribution than benchmark).
- **Actions:**
 - Created `scripts/calibrate_threshold_raw.py`.
 - Added `load_unsw_nb15_raw()` to handle 49-column, no-header format.
 - Calibrated threshold:
 - Optimal: 0.030684 (Recall 96.05%, FPR 51.55%)
 - Updated test script and reran accuracy verification.

3.2. Results & Decision

- **Findings:** Old threshold (0.027361) performed better overall (higher recall, lower FPR after Agent Two filtering).
 - **Recommendation:** Keep 0.027361 as Agent Two corrects most false alarms.
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4. ROC Curve Analysis & Visualization

- **Goal:** Select threshold using ROC analysis (maximize F1 or Youden's J statistic).
 - **Actions:**
 - Created `roc_threshold_analysis.py` for ROC/F1/Youden's J analysis.
 - Generated plots and CSVs for both raw and benchmark data.
 - **Results:**
 - Youden's J optimal threshold: ~0.18-0.23 (low FPR, recall ~60%)
 - Max F1 threshold: ~0.038 (high recall, high FPR)
 - Visualizations saved in [results](#).
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5.2. Training & Evaluation

- **Experiments:**
 - Trained ensemble with various contamination/nu values for IF/OCSVM.
 - Evaluated on raw and benchmark data.
- **Results:**
 - **Optimal config:** contamination=0.15, nu=0.15, AE threshold=0.0396

- **Performance:**

- Raw Data: Recall 76.9%, FPR 12.4% (vs. AE alone: Recall 98.65%, FPR 49.38%)
- Benchmark: Recall 79.6%, FPR 37.3% (vs. AE alone: Recall 99.06%, FPR 76.3%)